Layer	% Variance Explained	L1	L0	% Alive	CE Diff	% CE Recovered
Attn 0	92%	8	128	17%	0.02	99%
Attn 1	87%	9	127	17%	0.03	94%
Attn 2	90%	19	215	12%	0.05	93%
Attn 3	89%	12	169	13%	0.03	93%
Attn 4	83%	8	132	14%	0.01	95%
Attn 5	89%	11	144	20%	0.02	93%

Table 4: Attention SAE evaluation by layer.

Layer	% Variance Explained	L1	L0	% Alive	CE Diff	% CE Recovered
MLP 0	97%	5	5	40%	0.10	99%
MLP 1	85%	8	69	44%	0.06	95%
MLP 2	99%	12	88	31%	0.11	88%
MLP 3	88%	20	160	25%	0.12	94%
MLP 4	92%	20	100	29%	0.14	90%
MLP 5	96%	31	102	35%	0.15	97%

Table 5: MLP SAE evaluation by layer.

There exist multiple SAEs for every submodule. The primary difference between them is their average L0 norm.[3] Neuronpedia (Lin & Bloom, 2023) uses the SAEs with the L0 norm closest to 100; we do the same.

One of the primary technical challenges in using the Gemma Scope SAEs is the existence of BOS features. These are features that are active primarily or only on BOS tokens, and whose top logits are generally not informative. These features are difficult to interpret, but can have high indirect effects on the model's logits. As we cannot interpret them, we exclude them from annotation and from the SHIFT analysis (i.e., we do not ablate them). We also exclude them when running the feature skyline in §4.

C Feature Circuits

C.1 Subject-Verb Agreement

Here, we present the full agreement circuits for various syntactic agreement structures, with researcher-provided annotations for features. We chose thresholds manually in order to keep the number of nodes to annotate manageable while still displaying the full range of feature types for a given task.

In each circuit, sparse features are shown in rectangles, whereas causally relevant error terms not yet captured by our SAEs are shown in triangles. Nodes shaded in darker colors have stronger effects on the target metric m. Blue nodes and edges are those which have positive indirect effects (i.e., are useful for performing the task correctly), whereas red nodes and edges are those which have counterproductive effects on m (i.e., cause the model to consistently predict incorrect answers).

First, we present agreement across a relative clause. Pythia (Figure 7) and Gemma (Figure 8) both appear to detect the subject's grammatical number at the subject position. One position later, features detect the presence of relative pronouns (the start of the distractor clause). Finally, at the last token of the relative clause, the attention moves the subject information to the last position, where it assists in predicting the correct verb inflection. Gemma 2 additionally leverages noun phrase (NP) number tracking features, which are active at all positions for NPs of a given number (except on distractor phrases of opposite number). We present an example of an NP number tracker feature in Figure 15.

The circuits for agreement across a prepositional phrase (Figures 9 and 10) look remarkably similar to agreement across a relative clause; for both Pythia and Gemma, these two circuits share over 85%

[3] In other words, the average number of features active for a given token.

Layer	% Variance Explained	L1	L0	% Alive	CE Diff	% CE Recovered
Resid 0	92%	11	59	41%	0.24	97%
Resid 1	85%	13	54	38%	0.45	95%
Resid 2	96%	24	108	27%	0.55	94%
Resid 3	96%	23	68	22%	0.58	95%
Resid 4	88%	23	61	27%	0.48	95%
Resid 5	90%	35	72	45%	0.55	92%

Table 6: Residual (Resid) SAE evaluation by layer.

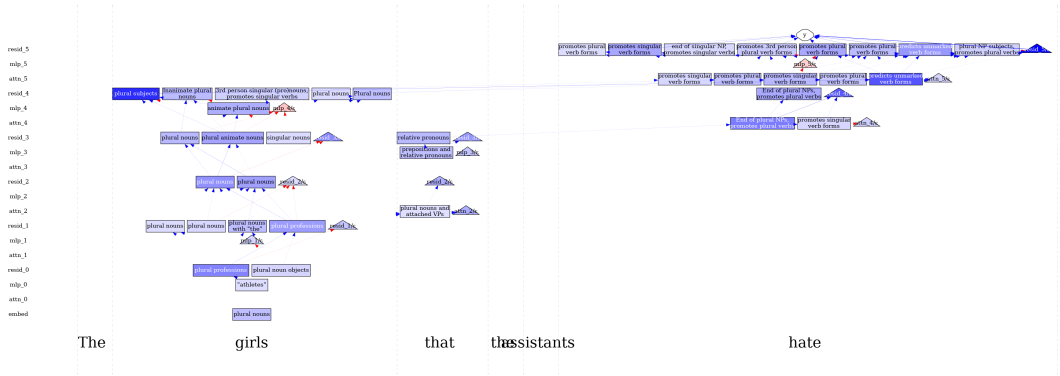

Figure 7: The feature circuit for agreement across a relative clause in Pythia-70M, computed using $T_N = 0.1$ and $T_E = 0.01$. The model detects the subject's number at the subject position. Other features detect relative pronouns (the start of the distractor clause). Finally, at the last token of the RC, the attention moves the subject information to the last position, where it assists in predicting the correct verb inflection.

of their features, and many of the same features are used for detecting both prepositions and relative clauses.

For simple agreement (Figures 11 and 12), many of the same features that were implicated in noun number detection and verb number prediction in the previous circuits also appear here. The models detect the subject's number at the subject position in early layers. In later layers, these noun number detectors become inputs to verb number promoters, which activate on anything predictive of particular verb inflections.

The circuits for agreement within a relative clause (Figures 13 and 14) appear to have the same structure as that for simple agreement: subject number detectors in early layers, followed by verb number promoters in later layers.

C.2 Bias in Bios Circuit

Here, we present the full annotated circuit discovered for the Bias in Bios classifier trained on Pythia-70M (described in §4 and App. E). The circuit was discovered using $T_N = 0.1$ and $T_E = 0.01$. We observe that the circuit (Figure 16) contains many nodes which simply detect the presence of gendered pronouns or gendered names. A few features attend to profession information, including one which activates on words related to nursing, and another which activates on passages relating to science and academia.

C.3 Cluster Circuits

Here, we present full annotated circuits discovered for automatically discovered behaviors (described in App. G). First, we present the circuit for incrementing number sequences (Figure 17), discovered with $T_N = 0.4$ and $T_E = 0.04$. We note that this circuit includes many features which perform either succession (Gould et al., 2023) or induction (Olsson et al., 2022). The succession

Figure 8: The feature circuit for agreement across a relative clause in Gemma-2-2B, computed using $T_N = 0.073$ and $T_E = 0.007$.